



Informfully News Framework

Creating and Assessing News Recommender Systems

Team University of Zurich



**Universität
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Department of Informatics – Dynamic and Distributed Information Systems Group



Setting the Scene



Workshop Agenda

- Half-day of a workshop on diversity in news recommender systems
- Interdisciplinary audience (programming skills cannot be assumed)
- Program overview for the workshop:
 - 10:00-10:30 Introduction and presentation
 - 10:30-12:15 Breakout groups (3 groups and 3 tables, 30' per table + 15' break)
 - 12:15-13:00 Presentation of group results (15') Panel discussion (30')



TreeTalk Agenda

- Program overview for the TreeTalk:
 - 10' presentation
 - 10' notebook
 - 10' feedback
- What we want from you:
 - Let us know if the overview we provide is lacking something or hard to understand.
 - Please have 2-3 point ready for the feedback round.



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Introduction



Informfully News App

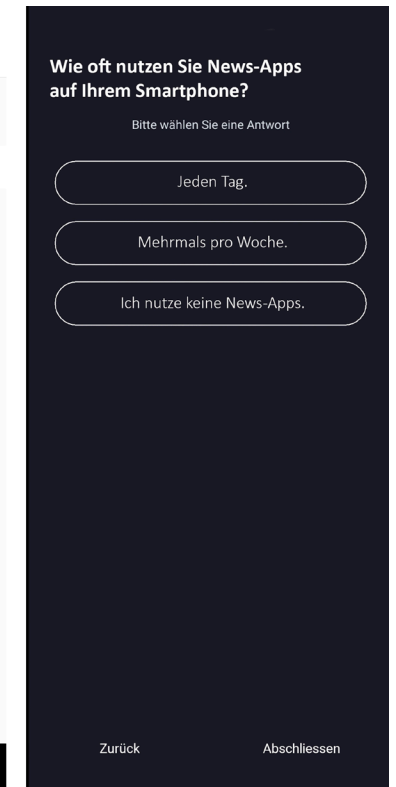
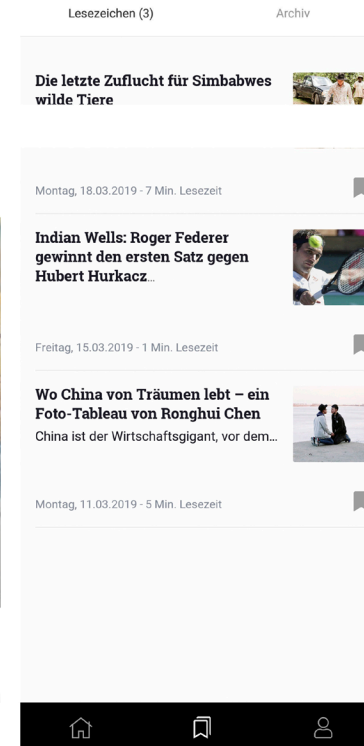
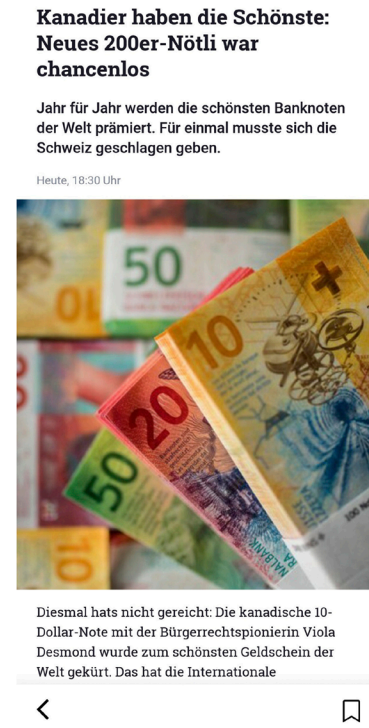
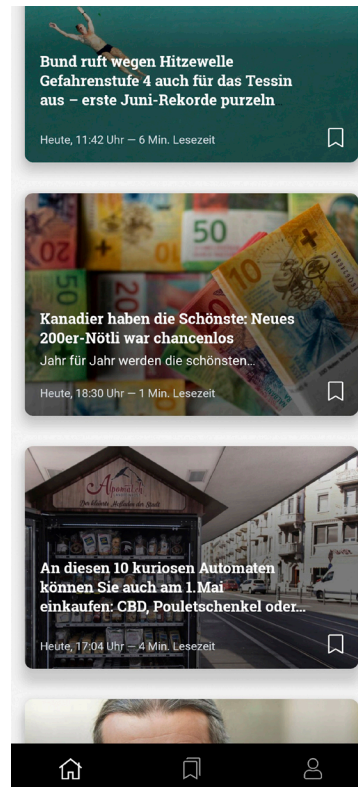




Informfully News App

Username: test@test.test

Password: test





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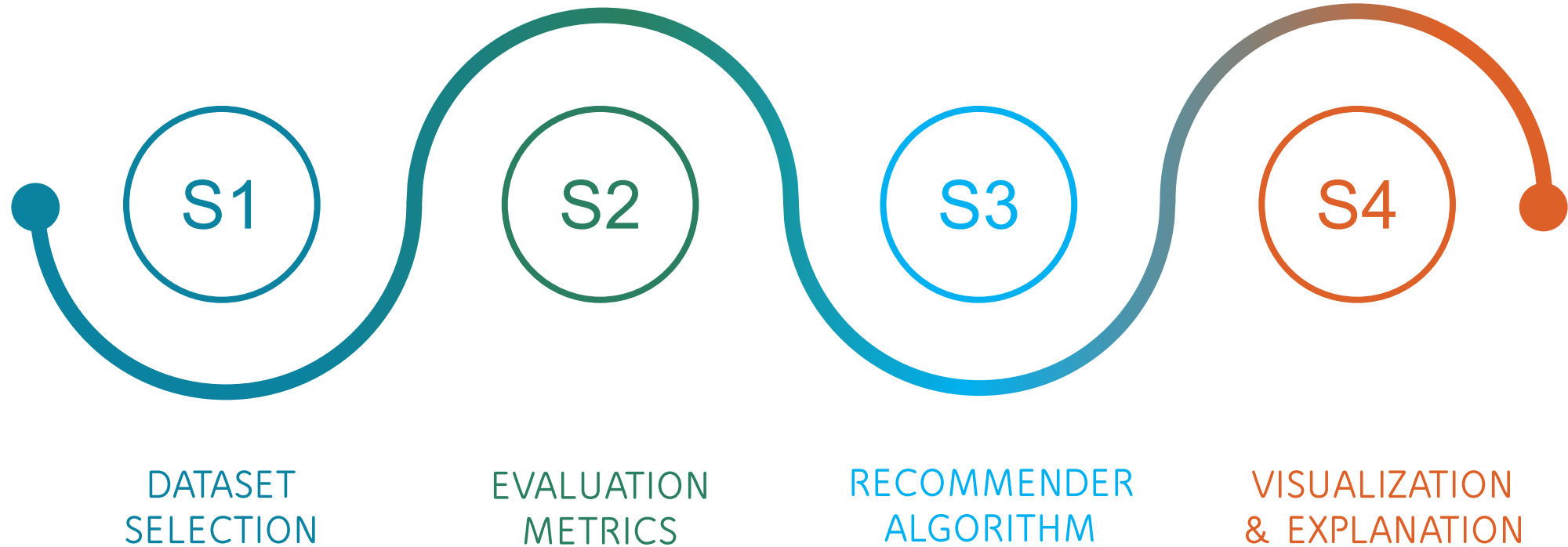
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Recommender Framework

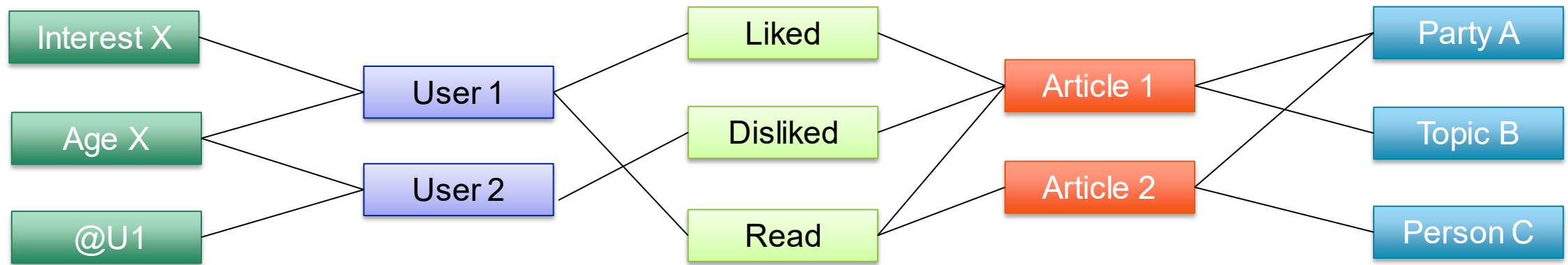


4-Step Framework for Recommender Systems





1. Dataset Selection



The breakout group session highlights:

- MINDS dataset (user and article data, articles with topic labels)
- UZH CH dataset (user and article data, articles with political labels and user ratings)
- UHH/UZH DE dataset (user article data, articles with tagged parties/politicians and user ratings)



2. Evaluation Metrics

- Dataset selection determines:
 - The scope of the evaluation (the metrics that are recorded and assessed)
 - The mode of the evaluation (offline or online mode)
- The main challenge for evaluation metrics is the goal selection!

The breakout group session highlights:

- Examples of survey evaluation on user satisfaction
- User engagement rates for automated assessment
- Implicit and explicit user feedback for satisfaction ratings

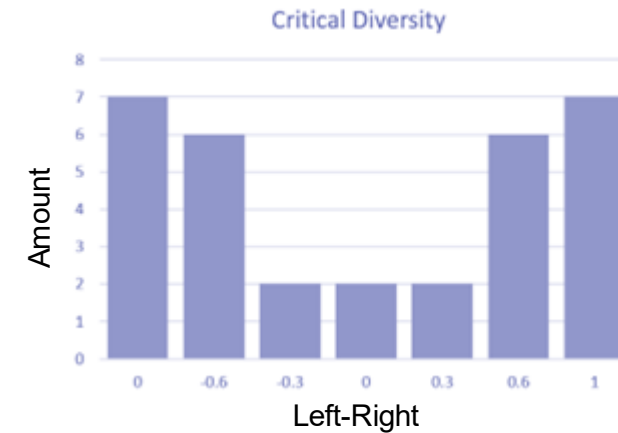
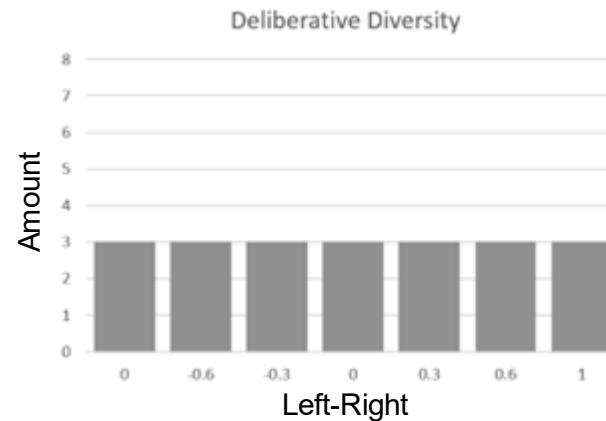
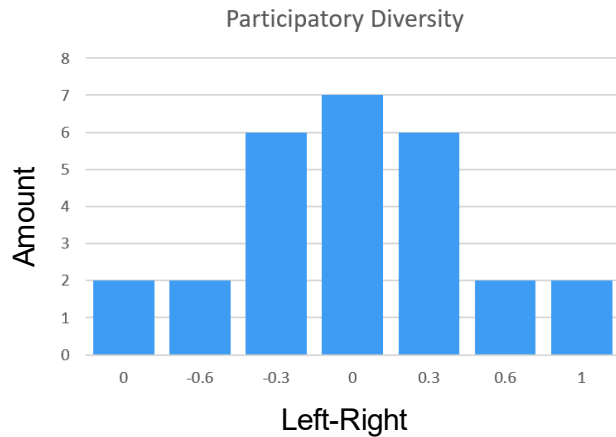


3. Recommender Algorithms

- Dataset selection determines:
 - Item-based approach (item data is available)
 - User-based approach (user data is available)
 - Hybrid approach (both user and item data is available)
- Evaluation metrics determine:
 - What and how to record user interaction
 - The overall goal of the recommender system
- Main challenge for the recommender algorithm part is to operationalize this goal!



3. Recommender Algorithms



The breakout group session highlights:

- Diversity is always **normative** and there exist **multiple** definitions
- Exposure diversity (UHH/UZH DE news dataset)
- Political diversity (UZH CH news dataset)



4. Visualization and Explanations

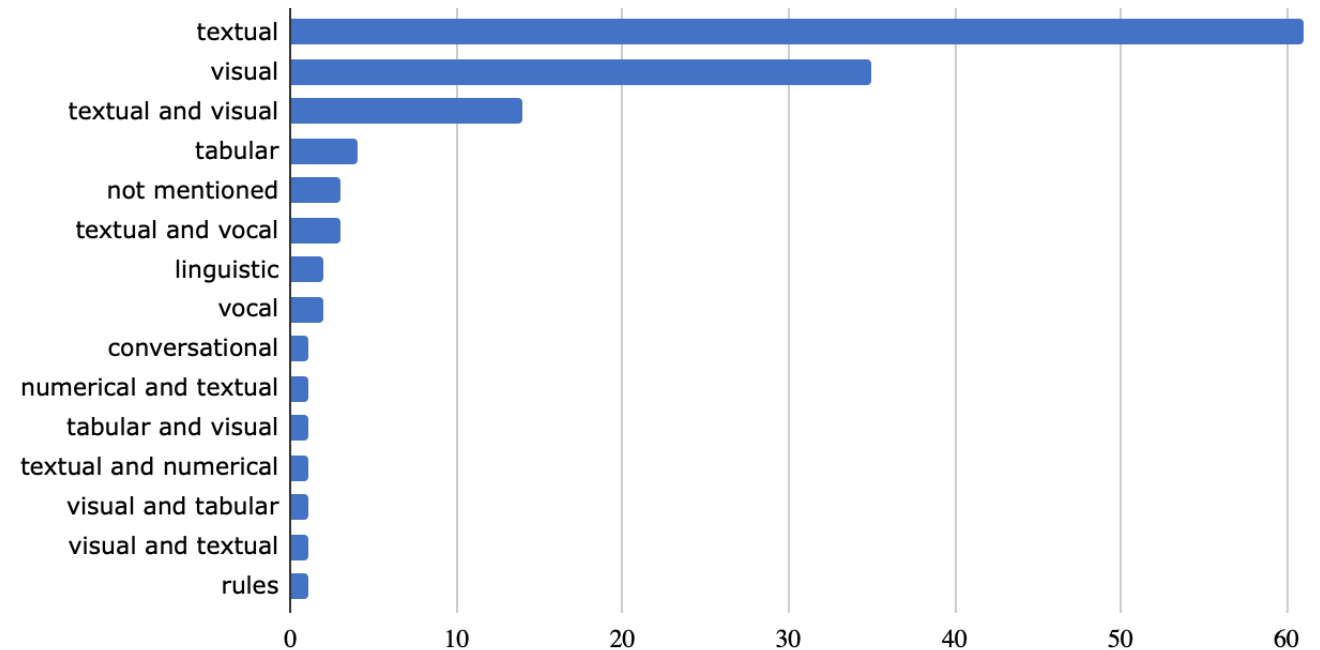
- Dataset selection determines what is shown to the users
- Evaluation metrics determine what tools (e.g., rating features) need to be provided
- Algorithms determine the order items are shown to users
- Main challenge for visualization is the presentation of recommendations to users!



4. Visualization and Explanations

Step 1: What form of explanation?

- What explanation do we need?
- What do user require?
- What does the law require?

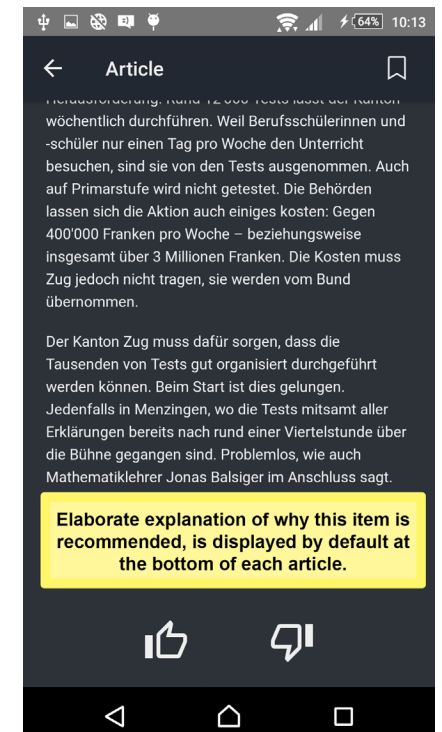
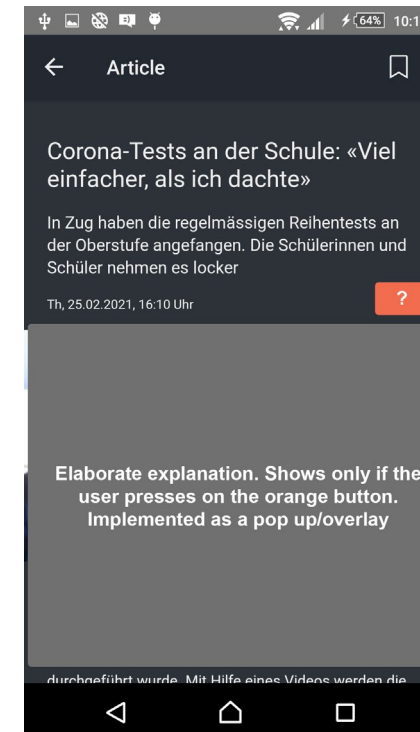
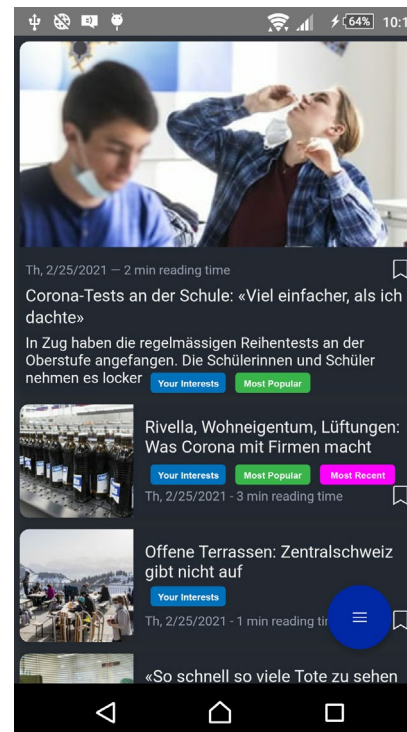




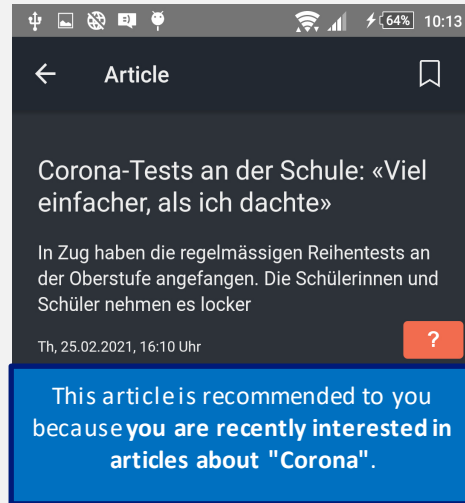
4. Visualization and Explanations

Step 2: How and what to explain?

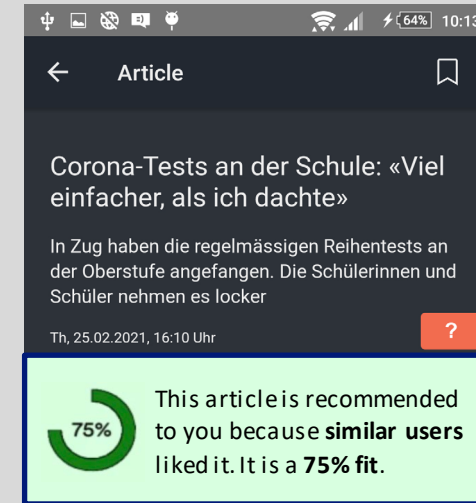
- How to visualize explanations?
- What is the content of explanations?
- Are explanations personalized?
- What are user's information needs?
- How much details are needed?



User Preferences & Input



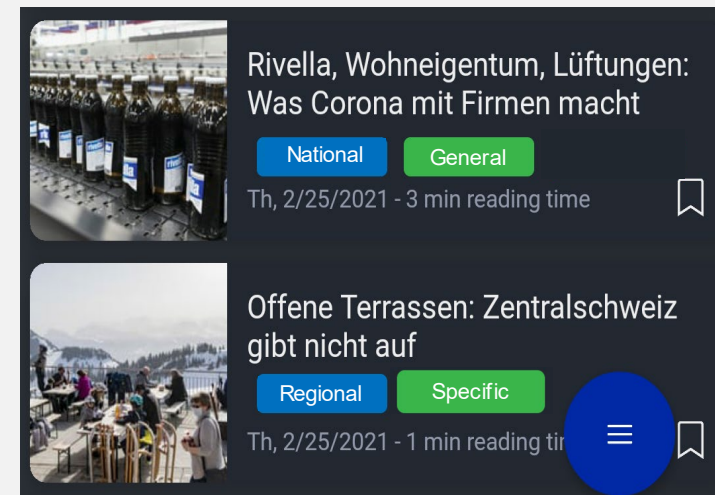
Decision Inference Process



Background & Complementary Information



Alternatives and their Features





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Breakout Group Session



Organization and Instructions

- Think of a use case and form groups accordingly (e.g., reading news on websites)
- Split in 3 groups and pick a table (rotation every 30')
- Write down your finding for the discussion
- Table overview
 - Table 1: Datasets and evaluation metrics (Oana)
 - Table 2: Recommender algorithms (Lucien)
 - Table 3: Visualization and explanations (Athina & Kathrin)



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Group Presentations and Discussion Round



Group Results

- What was your use case?
- How did you tackle the following aspects:
 - Dataset selection
 - Evaluation metrics
 - Recommender algorithms
 - Visualization & explanations
- Feedback questions:
 - Did the framework shed light on a problematic aspect?
 - What feature was most beneficial, what was lacking?



Discussion Round (TreeTalk Version)

- Have you ever participated in an interdisciplinary discussion/workshop?
- How did you discuss technical details (code etc.) in that setting?
- What other hand-on components could you imagine for explaining rec. systems?
- Is there a good rec. system you want us to include and feature?



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Thank you very much for your participation!